INKLOOM AI HACKATHON | ROUND 2: THE BUILD | WORKING PROTOTYPE SUBMISSION

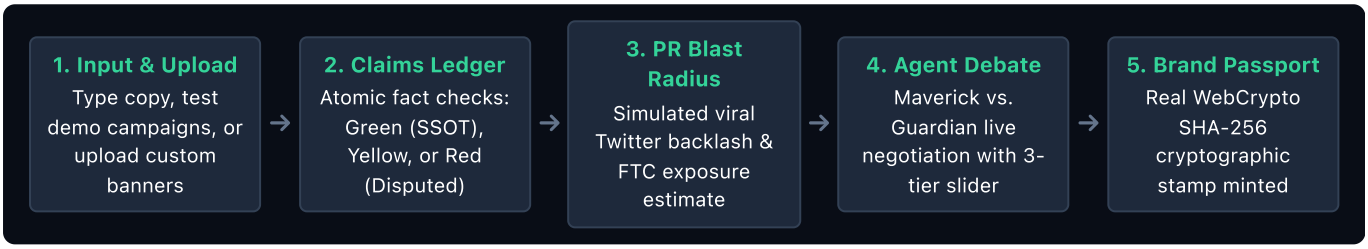
# BrandCircuit

Multimodal Pre-Flight CI/CD & Claims Ledger for Enterprise Brand Content

| LIVE REPOSITORY | CORE WORKFLOW | ARCHITECTURE | EVALUATION WINDOW |
|---|---|---|---|
| github.com/SrinivasaKamathB/BrandCircuit | 100% Testable & Functional | WebCrypto SHA-256 + Node.js 24 | 22–25 October 2026 |

## 1 Working Core Workflow (Demonstrable Beginning-to-End)

**BrandCircuit** transforms brand governance from a bureaucratic bottleneck into an automated pre-flight CI/CD pipeline. The Round 2 build is fully interactive, featuring both 1-click test datasets and custom manual input workflows:




## 2 List of Completed Features (Round 2 MVP)

| Feature Module | Technical Implementation | User Capability |
|---|---|---|
| **Verified Claims Ledger (SSOT)** | Deconstructs drafts into atomic factual statements. Cross-references against internal corporate Single Source of Truth into Verified, Needs Citation, and Disputed. | Inline color-coded badges with direct 1-click "Resolve" routing. |
| **PR Blast Radius & Crisis Simulator** | Calculates potential financial exposure under FTC §5 and renders a simulated public backlash thread with community notes and skeptical investor replies. | Visualizes real-time business and legal risk before publication. |
| **Maverick vs. Guardian Agent Debate** | Multi-agent negotiation between Growth Marketing (high conversion) and Compliance Counsel (zero liability) with quick-click presets (Strict, Balanced, High Growth). | Compromise engine dynamically updates copy while preserving creative punch. |
| **Multimodal Visual DNA Inspector** | Client-side canvas inspector analyzing marketing banners for logo clear-space exclusion boundaries, color drift tolerance ($\Delta E = 4.2$), and WCAG contrast. | Supports real image uploads with live thumbnail and canvas overlay. |
| **Cryptographic Brand Passport** | Uses browser WebCrypto API (`crypto.subtle.digest`) to generate authentic SHA-256 digests once content satisfies all compliance checks. | Tamper-proof digital seal allowing downstream CMS and social scheduling. |
| **Role-Based Workspaces** | Instant toggle between Creator Drafting Mode and Reviewer Sign-Off Queue. | Simulates production handover between agencies and compliance leads. |

## 3 Technology & Implementation Details

### 💻 Frontend Architecture

- **Responsive Cyber-Themed UI:** High-contrast dark aesthetic with Tailwind CSS, custom glowing badges, and Lucide/FontAwesome iconography.
- **Dynamic Policy Engine:** Active text scanner that analyzes custom copy for prohibited guarantees,

### ⚙️ Backend & Security Architecture

- **Node.js v24 Microserver (`server.js`):** Lightweight, zero-dependency async HTTP server ensuring 100% portability.
- **Real WebCrypto API Cryptography:** Mathematically valid SHA-256 content hashing for tamper detection.

unverified claims, and aggressive tone.
- **HTML5 Canvas Visual Pipeline:** Interactive image inspector rendering logo boundaries and color drift analysis.

- **Enterprise Hardening:** Production OWASP headers including strict CSP, X-Frame-Options: DENY, nosniff, and zero data retention.

## 4 Explanation of Changes Made to Original Plan

During Round 1, our conceptual proposal focused on static text highlighting and standard handbook lookups. In building the Round 2 MVP, we implemented three deliberate architectural evolutions:

- **From Passive Warnings to "PR Blast Radius":** Simple red warning banners fail to motivate marketing teams. We introduced the *PR Blast Radius & Crisis Simulator* to visualize the concrete cost of failure (viral backlash and FTC regulatory exposure), creating an immediate executive business case.
- **From Rigid LLM Overrides to "Multi-Agent Negotiation":** Copywriters reject standard AI rewrites because they remove creative voice and conversion punch. We introduced the *Maverick vs. Guardian Debate* with a 3-tier compromise slider, enabling creators to balance legal safety with growth hooks.
- **From Status Flags to "Cryptographic Passports":** Rather than a superficial "Approved" checkbox, we engineered a verifiable SHA-256 digital passport to prevent post-approval content tampering in enterprise publishing pipelines.

## 5 Three Major Improvements for Round 3 (The Mastermind Roadmap)

As mandated by competition rules, these three meaningful additions represent deep architectural expansions:

IMPROVEMENT 1 **Self-Healing Playbook**

**Autonomous Git & Figma PRs:** When the system detects repeated team blindspots across 50+ posts, it autonomously creates GitHub Pull Requests updating `brand-guidelines.md` and syncs new design tokens directly into Figma. The brand handbook continuously heals and updates itself.

IMPROVEMENT 2 **10-Persona Jury**

**Synthetic Audience Simulation:** Expands the 2-agent debate into an autonomous panel of 10 synthetic AI personas (FTC auditor, skeptical enterprise buyer, competitor PR lead, tech critic) to predict virality, controversy, and sentiment with statistical confidence bounds before ad spend.

IMPROVEMENT 3 **ZK Brand Proofs**

**Zero-Knowledge Publishing Mesh:** Issues cryptographic proofs allowing enterprises to verify compliance with external agencies and ad platforms without exposing confidential embargoed launch data, with direct automated publishing to CMS and social networks.

## 6 Quick-Start Instructions for Evaluators

| Step | Action | What to Observe |
|---|---|---|
| **1. Launch App** | Run `node server.js` and open `http://localhost:3000`. | Cyber-themed dark interface with live system metrics. |
| **2. Load Demo** | Click **"Load Risky Campaign (Violations)"** at top right. | Compliance drops to 64%; PR Blast Radius spikes to 87%. |
| **3. Resolve Claims** | Click **"Resolve"** in the Claims Ledger $\rightarrow$ pick **"Balanced"** $\rightarrow$ click **"Apply Fix"**. | Draft auto-updates; Compliance jumps to 96% (Green). |
| **4. Visual Check** | Click **"Upload Banner"** or use sample $\rightarrow$ click **"Inspect DNA"**. | Canvas overlay checking logo padding & hex color drift. |
| **5. Certify** | Click **"Certify & Mint Cryptographic Passport"**. | Real SHA-256 mathematical hash generated for publishing. |